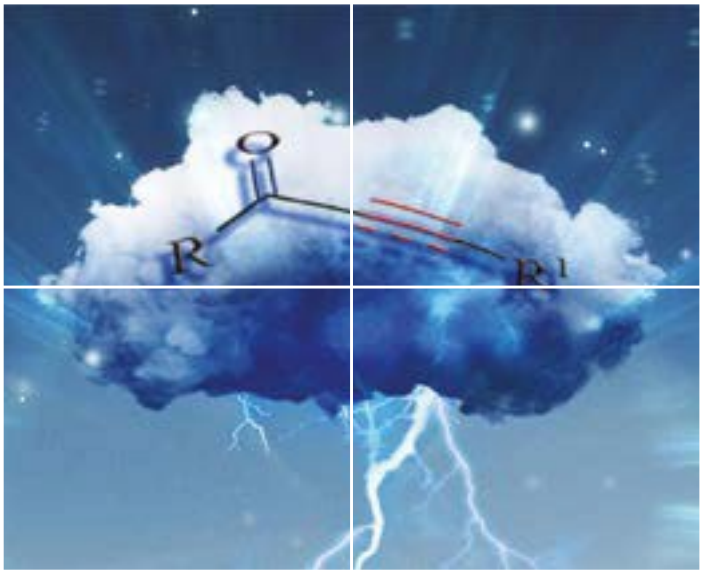
Check for updates

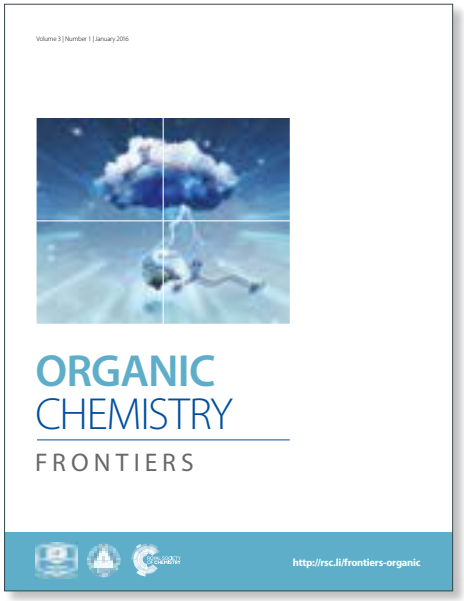
View Article Online
View Journal

ORGANIC CHEMISTRY
FRONTIERS



Accepted Manuscript

This article can be cited before page numbers have been issued, to do this please use: Y. Han and S. Ma, *Org. Chem. Front.*, 2018, DOI: 10.1039/C8QO00650D.



This is an Accepted Manuscript, which has been through the Royal Society of Chemistry peer review process and has been accepted for publication.

Accepted Manuscripts are published online shortly after acceptance, before technical editing, formatting and proof reading. Using this free service, authors can make their results available to the community, in citable form, before we publish the edited article. We will replace this Accepted Manuscript with the edited and formatted Advance Article as soon as it is available.

You can find more information about Accepted Manuscripts in the author guidelines.

Please note that technical editing may introduce minor changes to the text and/or graphics, which may alter content. The journal's standard Terms & Conditions and the ethical guidelines, outlined in our author and reviewer resource centre, still apply. In no event shall the Royal Society of Chemistry be held responsible for any errors or omissions in this Accepted Manuscript or any consequences arising from the use of any information it contains.

Please do not adjust margins
View Article Online
DOI: 10.1039/C8QO00650D

Journal Name

Research Article

Rhodium-catalyzed Highly Diastereoselective Intramolecular [4+2] Cycloaddition of 1,3-Disubstituted Allene-1,3-dienes

Received 00th January 20xx,
Accepted 00th January 20xx

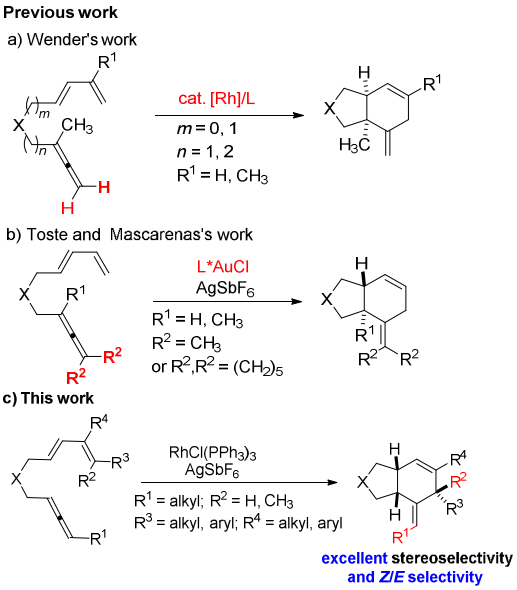
DOI: 10.1039/x0xx00000x

www.rsc.org/

Yulin Han[a,b] and Shengming Ma*[a, c]

$RhCl(PPh_3)_3$-catalyzed [4+2] intramolecular cycloaddition of allene-1,3-dienes afforded *cis*-fused [3.4.0]-bicyclic products with three chiral centers in good yields with an excellent chemo- and diastereoselectivity. The configuration of the C=C bonds in the 1,3-diene unit controls the relative configurations of the non-bridging tertiary carbon atom in the six-membered ring. Based on the experimental results, a mechanism involving cyclometalation has been proposed.

Cycloaddition reactions are a practical and efficient method to construct cyclic products from simple acyclic starting materials by generating at least two bonds and one ring in only one step.[1] For example, [4+2] cycloaddition between 1,3-dienes and alkenes or alkynes is one of the most powerful synthetic tools to assemble six-membered cyclic structures.[2-4] In 1995, Wender and coworkers reported the intramolecular [4+2] cycloaddition reaction of disubstituted terminal allenes with 1,3-dienes to afford *cis*-6,6-fused rings or *cis*-6,5-fused rings with nickel or rhodium complex as catalyst (Scheme 1, a).[5] The chemoselectivity mainly depends on the length of the tether between the diene and terminal allene. In 2009, the groups of Toste and Mascareñas independently demonstrated the gold-catalyzed intramolecular [4+2] cycloaddition reaction of symmetric tetrasubstituted allene with 1,3-diene to give *trans*-6,5-fused rings (Scheme 1, b).[6] So far, there is very limited report on applying 1,3-disubstituted allene unit as the dienophile.[7] The challenge is the *Z*/E selectivity of the remaining C=C bond as well as the diastereoselectivity of the three chiral centers generated during the reaction (Scheme 1, c).[8] Recently, there have been many advances in the development of new synthetic approaches to prepare 1,3-disubstituted allenes.[9,10] Herein, we present the $RhCl(PPh_3)_3$-catalyzed intramolecular [4+2] cycloaddition of 1,3-disubstituted allene-dienes, during which the internal C=C bond of the allene moiety reacted with the 1,3-diene unit to produce *cis*-6,5-fused bicyclic products with an excellent *Z*/*E* selectivity for the remaining C=C bond and an excellent diastereoselectivity referring to the three chiral centers (Scheme 1, c).



Scheme 1. Metal-catalyzed intramolecular [4+2] cycloaddition of allene-1,3-dienes.

We started our study on the reaction with 1,3-disubstituted allene-1,3-diene **1a** as the model substrate. The desired product was not formed under the catalysis of $Ni(COD)_2$ or $[Rh(COD)_2Cl]_2$ (entries 1 and 2, Table 1). The cationic rhodium catalyst $[Rh(COD)_2]BF_4$ also couldn't catalyze the reaction either (entry 3, Table 1). Interestingly, when $[RhCl(CO)_2]_2$ was used, the reaction processed smoothly to afford the expected bicyclic product **1a** in 81% yield with a *Z*/*E* ratio of 50:50 (entry 4, Table 1). The structure and relative configurations of *Z*-**2a**

[a] *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, P. R. China.* E-mail: masm@sioc.ac.cn
[b] *University of Chinese Academy of Sciences, Beijing 100049, P. R. China.*
[c] *Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, 220 Handan Road, Shanghai 200433, P. R. China.*

Electronic supplementary information (ESI) available: Preparation and characterisation data as well as ^1H and ^{13}C NMR spectra of all compounds. See DOI: 10.1039/x0xx00000x

Published on 31 July 2018. Downloaded by Kaohsiung Medical University on 7/31/2018 7:16:44 AM.

Organic Chemistry Frontiers Accepted Manuscript

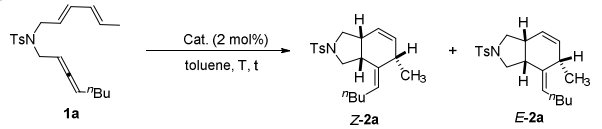
View Article Online
DOI: 10.1039/C8QO00650D

Published on 31 July 2018. Downloaded by Kaohsiung Medical University on 7/31/2018 7:16:44 AM.

and *E*-**2a** were established via analogy with the X-ray single crystal diffraction study of **2c** (Figure 1)[11] and the NOESY spectra (see the supporting information). The catalyst $[(C_{10}H_8)Rh(COD)]^+SbF_6^-$ used in Trost's report[7a] can facilitate the reaction to afford the cycloaddition product *Z*-**2a** with an incomplete conversion (entry 5, Table 1). To our delight, the yield was improved to 90% when $RhCl(CO)(PPh_3)_2$ was used with the exclusive formation of *Z*-**2a** (entry 6, Table 1). With $RhCl(PPh_3)_3$ as the catalyst, the yield was improved to 93% (entry 7, Table 1). The desired product could also be obtained in 91% yield at a lower temperature of 40 °C (entries 8 and 9, Table 1). The yield was 95% when running the reaction at a concentration of 0.2 M (entry 10, Table 1). The yield dropped to 85% with 1 mol% $RhCl(PPh_3)_3$ (entry 11, Table 1). Thus, the reaction conducted with **1a** and $RhCl(PPh_3)_3$ (2 mol%) in toluene at 40 °C was chosen as the standard conditions for further study.

With the optimized conditions in hand, the scope of substrates was investigated. The substrates with NTs as the linker were investigated firstly. The reaction afforded corresponding product *Z*-**2a**-**2c** in decent yields when R^1 is methyl, *n*-butyl, and cyclohexyl (entries 1-4, Table 2). The relative configurations in **2c** were established by the X-ray single crystal diffraction study[11] (Figure 1)-all three H atoms are *cis* oriented with a *Z*-C=C bond. R^2 may be H, methyl, and *n*-propyl (entries 1-4, and 10, Table 2). The substrates with R^2 being aryl groups in the terminal of the diene moiety could afford the target products under the standard conditions with lower yields. The yields were improved by conducting the reactions at 80 °C (entries 5-7, Table 2). With a substituent such as *o*-Br, the reactions only afforded **2h** in 15% yield with 36% recovery of **1h** even at 80 °C for 36 h (entry 8, Table 2). To our delight, 66% yield of **2h** was obtained when the reaction was conducted with the extra addition of $AgSbF_6$ (3 mol%) (entry 9, Table 2). R^3 may be H or alkyl. It is worth mentioning that the reactions produced products **2a**-**2h** as the only diastereoisomer while in entry 10 of Table 2 a minor amount of other diastereoisomer was formed. Malonate could also be used as the tether of the substrates. However, the reaction of **1j** became sluggish as compared to the substrates with NTs as the tether and the corresponding product **2j** was afforded with the help of $AgSbF_6$ (entry 11, Table 2).

Table 1 Optimization of rhodium-catalyzed intramolecular [4+2] cycloaddition of racemic allene-1,3-diene **1a**[a]

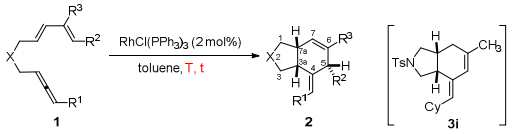


1a *Z*-**2a** *E*-**2a**

Entry	[Rh]	T (°C)	t (h)	Yield of **2a**[b]	*Z*/*E*	Recovery of **1a**[b]
1	$Ni(COD)_2$	80	12	N.D.	-	68
2	$[Rh(COD)Cl]_2$	80	12	N.D.	-	90
3	$[Rh(COD)_2]BF_4$	80	12	N.D.	-	77
4	$[RhCl(CO)_2]_2$	80	12	81	50:50	N.D.
5[c]	$[(C_{10}H_8)Rh(COD)]^+SbF_6^-$	rt	12	59	100:0	12
6	$RhCl(CO)(PPh_3)_2$	80	12	90	100:0	N.D.
7	$RhCl(PPh_3)_3$	80	12	93	100:0	N.D.
8	$RhCl(PPh_3)_3$	60	5	92	100:0	N.D.
9	$RhCl(PPh_3)_3$	40	3	91	100:0	N.D.
10[d]	$RhCl(PPh_3)_3$	40	3	95	100:0	N.D.
11[e]	$RhCl(PPh_3)_3$	40	10	85	100:0	N.D.

[a] The reaction was conducted with **1a** (0.2 mmol) and catalyst (2 mol%) in 2 mL of toluene. [b] Determined by ^1H NMR analysis with mesitylene as the internal standard and N. D. = not detected. [c] The reaction was conducted with **1a** (0.1 mmol) and catalyst (3 mol%) in 2 mL of 1,2-dichloroethane. [d] 1 mL of toluene was used. [e] 1 mol% $RhCl(PPh_3)_3$ and 1 mL of toluene were used.

Table 2 The scope of rhodium-catalyzed intramolecular [4+2] cycloaddition of racemic allene-1,3-dienes [a]



1 **2** **3i**

Entry	X/R^1/R^2/R^3 (**1**)	T (°C)	T (h)	**2** Isolated yield (%)
1	NTs/^{n}Bu/CH_3/H ((2*E*,4*E*)-**1a**)	40	3	81 ((3a*R**,5*S**,7a*R**,*Z*)-**2a**)
2	NTs/CH_3/CH_3/H ((2*E*,4*E*)-**1b**)	40	8	82 ((3a*R**,5*S**,7a*R**,*Z*)-**2b**)
3	NTs/Cy/CH_3/H ((2*E*,4*E*)-**1c**)	40	3	73 ((3a*R**,5*S**,7a*R**,*Z*)-**2c**)
4	NTs/^{n}Bu/*n*-C_3H_7/H ((2*E*,4*E*)-**1d**)	40	4	78 ((3a*R**,5*S**,7a*R**,*Z*)-**2d**)
5[b]	NTs/CH_3/Ph/H ((2*E*,4*E*)-**1e**)	80	24	39 ((3a*R**,5*R**,7a*R**,*Z*)-**2e**)
6[b]	NTs/CH_3/*p*-FC_6H_4/H ((2*E*,4*E*)-**1f**)	80	21.5	71 ((3a*R**,5*R**,7a*R**,*Z*)-**2f**)
7[b]	NTs/CH_3/*p*-ClC_6H_4/H ((2*E*,4*E*)-**1g**)	80	15	63 ((3a*R**,5*R**,7a*R**,*Z*)-**2g**)
8[b]	NTs/^{n}Bu/*o*-BrC_6H_4/H ((2*E*,4*E*)-**1h**)	80	36	15[c] ((3a*R**,5*S**,7a*R**,*Z*)-**2h**)
9[d]	NTs/^{n}Bu/*o*-BrC_6H_4/H ((2*E*,4*E*)-**1h**)	80	8	66 ((3a*R**,5*S**,7a*R**,*Z*)-**2h**)
10	NTs/Cy/H/CH_3 ((*E*)-**1i**)	40	4	87[e] ((3a*R**,7a*R**,*Z*)-**2i**)
11[d]	$C(CO_2Me)_2$/^{n}Bu/Ph/H ((2*E*,4*E*)-**1j**)	40	24	81 ((3a*R**,5*R**,7a*R**,*E*)-**2j**)

[a] Conditions A: the reaction was conducted with **1** (1.0 mmol) and $RhCl(PPh_3)_3$ (2 mol%) in 5 mL of toluene. [b] The reaction was conducted on 0.5 mmol scale. [c] 36% yield of **1h** was recovered and the yield of **2h** was determined by the NMR spectrum of the crude reaction mixture. [d] Conditions B: the reaction was conducted with **1** (0.5 mmol), $RhCl(PPh_3)_3$ (2 mol%), and $AgSbF_6$ (3 mol%)

Organic Chemistry Frontiers Accepted Manuscript

This journal is © The Royal Society of Chemistry 20xx

Please do not adjust margins
View Article Online
DOI: 10.1039/C8QO00650D
Journal Name COMMUNICATION

in 2.5 mL of toluene. [e] The ratios of **2i/3i** in the isolated product and the crude product were 27:1 and 15:1 which were determined by the ^{1}H NMR analysis of the isolated product and the crude product.

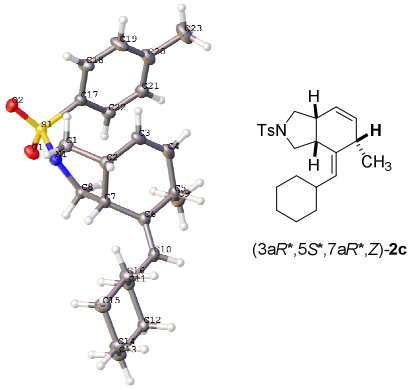
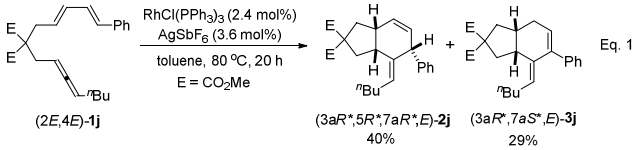


Figure 1 The ORTEP representations of (3a*R**,5*S**,7a*R**,*Z*)-**2c**.

When the phenyl-substituted 1,3-diene-incorporated substrate (2*E*,4*E*)-**1j** was reacted with $RhCl(PPh_3)_3$ and $AgSbF_6$ at 80 °C for 20 h, in addition to 40% yield of (3a*R**,5*R**,7a*R**,*E*)-**2j**, 29% yield of (3a*R**,7a*S**,*E*)-**3j** was obtained (Eq. 1). The relative configuration of (3a*R**,7a*S**,*E*)-**3j** was also established by the X-ray single crystal diffraction[12] (Figure 2)-the C=C bond in six-membered ring was conjugated with the exocyclic double bond.



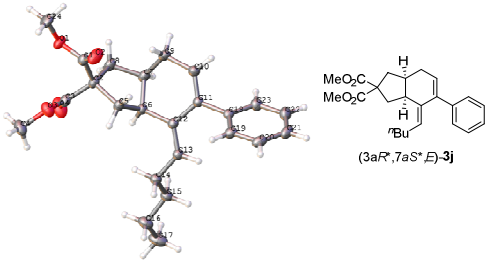
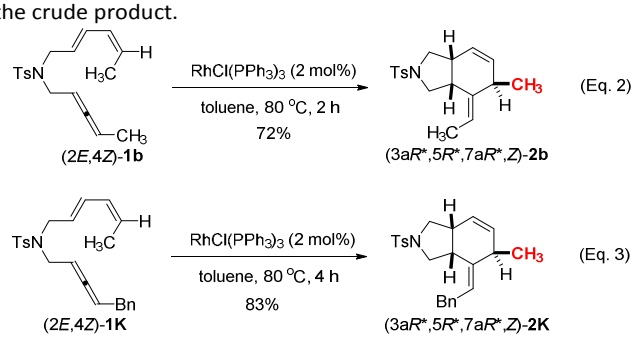


Figure 2 The ORTEP representations of (3a*R**,7a*S**,*E*)-**3j**.

In addition, when we applied the C=C bond stereoisomer (2*E*,4*Z*)-**1b** and **1k**, the reaction at 80 °C afforded the diastereoisomers (3*aR**,5*R**,7*aR**,*Z*)-**2b** and (3*aR**,5*R**,7*aR**,*Z*)-**2k** (Scheme 2). The relative configuration of (3*aR**,5*R**,7*aR**,*Z*)-**2b** was also established by the X-ray single crystal diffraction studies (Figure 3)[13]-the H atom of C5 in the six-membered ring is *trans* to the two H atoms of the bridged carbon atoms C3a and C7a. Thus, the configuration of the C5 was controlled by the *E*/*Z* configuration of the distal C=C bond in the diene moiety of the substrates. (Eqs. 2, 3 vs. entry 2, Table 2).



Scheme 2 Rhodium-catalyzed intramolecular [4+2] cycloaddition of (2*E*,4*Z*)-**1b** and (2*E*,4*Z*)-**1k**.

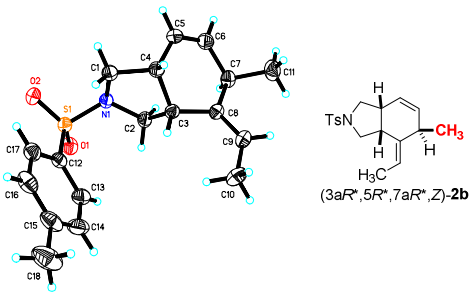


Figure 3 The ORTEP representation of (3*aR**,5*R**,7*aR**,*Z*)-**2b**.

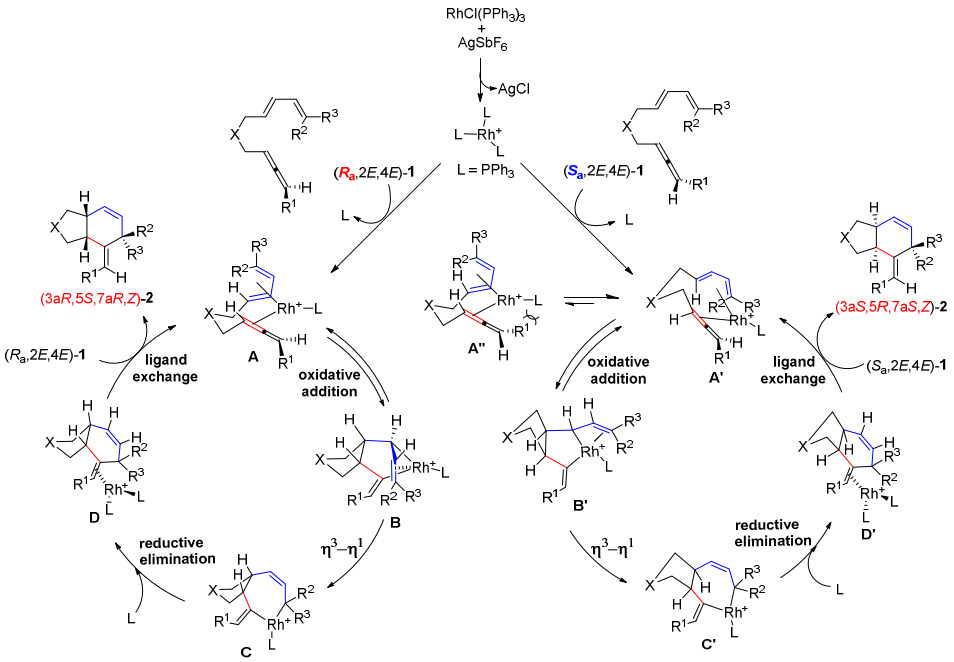
Based on the X-ray diffraction study of (3a*R**,5*S**,7a*R**,*Z*)-**2c**, (3*aR**,5*R**,7*aR**,*Z*)-**2b** and the observed diastereoselectivity, a mechanism is proposed (Scheme 3). Firstly, $RhCl(PPh_3)_3$ reacts with $AgSbF_6$ to generate the cationic catalyst $[Rh(PPh_3)_3]^+SbF_6^-$. For the reaction of *R*-isomer (R_a,2*E*,4*E*)-**1** with the Rh catalyst would generate Intermediate **A** , in which the "inner" C=C bonds in allene and the *s-cis*-1,3-diene coordinated with the rhodium atom in such a way that the two H atoms in intermediate **A** are *cis* orientated in order to make the R^1 group pointing away from the 1,3-diene unit. Subsequent cyclometalation leads to the formation of Intermediate **B.** The allylic rearrangement leads to the formation of rhodiabicycloheptene **C,** in which the three tertiary hydrogen atoms are *cis* orientated. Subsequent reductive elimination and ligand exchange with (R_a,2*E*,4*E*)-**1** provide the product (3a*R*,5*S*,7a*R*,*Z*)-**2** and regenerate the catalytically active species **A** to finish the catalytic cycle. For the *S*-enantiomer (S_a,2*E*,4*E*)-**1**, Intermediate **A'** other than Intermediate **A''** (which has the same coordination mode of Intermediate **A**) would be formed in order to keep the R^1 group far away from the catalyst. Subsequent cyclometalation, η^3-η^1 rearrangement, reductive elimination, and ligand exchange with (S_a,2*E*,4*E*)-**1** would afford the enantiomer (3a*S*,5*R*,7a*S*,*Z*)-**2**.

Published on 31 July 2018. Downloaded by Kaohsiung Medical University on 7/31/2018 7:16:44 AM.
Organic Chemistry Frontiers Accepted Manuscript

View Article Online
DOI: 10.1039/C8QO00650D

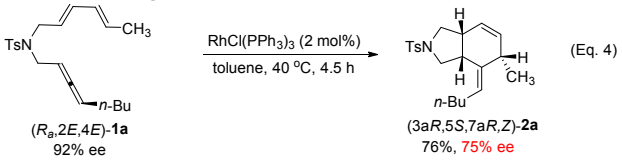
Journal Name

Research Article



Scheme 3 A proposed mechanism for the rhodium-catalyzed intramolecular [4+2] cycloaddition of allene-dienes (2*E*,4*E*)-**1** (R^3 ≠ aryl).

When the optically active allene-1,3-diene (*R*$_a$,2*E*,4*E*)-**1a** was conducted under the standard condition, only one diastereomer (3a*R*,5*S*,7a*R*,*Z*)-**2a** was obtained in 76% yield, unfortunately, the efficiency of chirality transformation was low and the ee of (3a*R*,5*S*,7a*R*,*Z*)-**2a** was only 75% (Eq. 4). We are still working on this issue.



(*R*$_a$,2*E*,4*E*)-**1a** 92% ee

(3a*R*,5*S*,7a*R*,*Z*)-**2a** 76%, 75% ee (Eq. 4)

In conclusion, we have developed the $RhCl(PPh_3)_3$-catalyzed intramolecular [4+2] cycloaddition of 1,3-disubstituted allene-1,3-dienes, giving *cis*-6,5-fused bicyclic products with a very high diastereoselectivity. In addition, the configurations in the C3a, C5, and C7a of the products are controlled by the configuration of the C=C bonds in the 1,3-diene unit in the starting materials. The three H atoms of the tertiary carbon in the product are *cis* to each other when (2*E*,4*E*)-1,3-disubstituted allene-dienes were used; the two *cis*-H atoms of the bridged carbon and the H atom of the other tertiary carbon in the six-membered ring are *trans* to each other if the (2*E*,4*Z*)-1,3-disubstituted allene-dienes were applied. A concerted cyclometalation, allylic rearrangement, reductive elimination has been proposed to account for the observed diastereoselectivity. Further studies of the chirality transfer of optically active allene-1,3-dienes are being conducted in our laboratory.

Acknowledgement

Financial support from National Natural Science Foundation of China (21690063) is greatly appreciated. We thank Mr Chaofan Huang in this group for reproducing the results of (3a*R**,5*R**,7a*R**,*Z*)-**2f,** (3a*R**,7a*R**,*Z*)-**2i** in Table 2 and (3a*R**,5*R**,7a*R**,*Z*)-**2b** in Scheme 2 presented in this study.

Notes and references

1 For selected reviews of cycloaddition, see: (a) M. Lautens, W. Klute and W. Tam, *Chem. Rev.* **1996**, *96*, 49. (b) K. V. Gothelf and K. A. Jørgensen. *Chem. Rev.* **1998**, *98*, 863. (c) I. Nakamura and Y. Yamamoto, *Chem. Rev.* **2004**, *104*, 2127. (d) M. Meldal and C. W. Tornøe, *Chem. Rev.* **2008**, *108*, 2952. (e) J. E. Hein and V. V. Fokin, *Chem. Soc. Rev.* **2010**, *39*, 1302. (f) W. Carruthers, *Cycloaddition Reactions in Organic Synthesis*; Pergamon: Oxford, 1990; pp 1-208.
2 For selected reviews of Diels-Alder reaction, see: (a) W. Oppolzer, In *Comprenhensive Organic Synthesis*, Vol. 5 (Eds.:

This journal is © The Royal Society of Chemistry 20xx

Published on 31 July 2018. Downloaded by Kaohsiung Medical University on 7/31/2018 7:16:44 AM.

Organic Chemistry Frontiers Accepted Manuscript

Please do not adjust margins
View Article Online
DOI: 10.1039/C8QO00650D

Published on 31 July 2018. Downloaded by Kaohsiung Medical University on 7/31/2018 7:16:44 AM.

B. M. Trost and I. Fleming), Pergamon Press: Oxford, 1991; pp 315-399. For selected reviews on applications of Diels-Alder reactions in total synthesis, see: (b) K. C. Nicolaou, S. A. Snyder, T. Montagnon and G. Vassilikogiannakis, *Angew. Chem. Int. Ed.* **2002**, *41*, 1668. (c) K. Takao, R. Munakata and K. Tadano, *Chem. Rev.* **2005**, *105*, 4779.

3 For representative examples with 1,3-dienes and alkenes, see: (a) R. S. Jolly, G. Luedtke, D. Sheehan and T. Livinghouse, *J. Am. Chem. Soc.* **1990**, *112*, 4965. (b) L. McKinstry and T. Livinghouse, *Tetrahedron*, **1994**, *50*, 6145. (c) D. F. Harvey and E. M. Grenzer, *J. Org. Chem.* **1996**, *61*, 159. (d) D. J. R. O'Mahony, D. B. Belanger and T. Livinghouse, *Synlett* **1998**, 443. (e) H. Heath, B. Wolfe, T. Livinghouse and S. K. Bae, *Synthesis* **2001**, 2341. (f) S. R. Gilbertson and G. S. Hoge, *Tetrahedron Lett.* **1998**, *39*, 2075. (g) S. R. Gilbertson, G. S. Hoge and D. G. Genov, *J. Org. Chem.* **1998**, *63*, 10077. (h) B. Wang, P. Cao and X. Zhang, *Tetrahedron Lett.* **2000**, *41*, 8041. (i) Y. Sato, Y. Oonishi and M. Mori, *Organometallics* **2003**, *22*, 30. (j) D. J. R. O'Mahony, D. B. Belanger and T. Livinghouse, *Org. Biomol. Chem.* **2003**, *1*, 2038.

4 For representative intramolecular examples with 1,3-dienes and alkynes, see: (a) P. A. Wender and T. E. Jenkins, *J. Am. Chem. Soc.* **1989**, *111*, 6432. (b) R. S. Jolly, G. Luedtke, D. Sheehan and T. Livinghouse, *J. Am. Chem. Soc.* **1990**, *112*, 4965. (c) P. A. Wender and T. E. Smith, *J.Org. Chem.* **1996**, *61*, 824. (d) K. Kumar and R. S. Jolly, *Tetrahedron Lett.* **1998**, 3047. (e) D. Motoda, H. Kinoshita, H. Shinokubo and K. Oshima, *Angew. Chem. Int. Ed.* **2004**, *43*, 1860. (f) A. Fürstner and C. C. Stimson, *Angew. Chem. Int. Ed.* **2007**, *46*, 8845. (g) H. Kusama, Y. Karibe, Y. Onizawa and N. Iwasawa, *Angew. Chem. Int. Ed.* **2010**, *49*, 4269. (h) S. M. Kim, J. H. Park and Y. K. Chung, *Chem. Commun.* **2011**, *47*, 6719. (i) K. Aikawa, S. Akutagawa and K. Mikami, *J. Am. Chem. Soc.* **2006**, *128*, 12648. (j) W.-J. Yoo, A. Allen, K. Villeneuve and W. Tam, *Org. Lett.* **2005**, *7*, 5853. For intermolecular examples with 1,3-dienes and alkynes, see: (k) A. Carbonaro, A. Greco, and G. Dall'Asta, *J. Org. Chem.* **1968**, *33*, 3948. (l) H. tom Dieck and R. Diercks, *Angew. Chem. Int. Ed. Engl.* **1983**, *22*, 778. (m) S.-J. Paik, S. U. Son and Y. K. Chung, *Org. Lett.* **1999**, *1*, 2045. (n) G. Hilt, W. Hess and K. Harms, *Org. Lett.* **2006**, *8*, 3287. For a related cycloaddition between unactivated vinylallenes and alkynes, see: (o) M. Murakami, M. Ubukata, K. Itami and Y. Ito, *Angew. Chem. Int. Ed.* **1998**, *37*, 2248. For representative examples on Hetero-Diels-Alder reactions, respectively, see: (p) B. M. Trost, R. E. Brown and F. D. Toste, *J. Am. Chem. Soc.* **2000**, *122*, 5877. (q) I. Koyama, T. Kurahashi and S. Matsubara, *J. Am. Chem. Soc.* **2009**, *131*, 1350.

5 (a) P. A. Wender, T. E. Jenkins and S. Suzuki, *J. Am. Chem. Soc.* **1995**, *117*, 1843. (b) P. A. Wender, M. P. Croatt and N. M. Deschamps, *Angew. Chem. Int. Ed.* **2006**, *45*, 2459. (c) B. C. Lainhart and E. J. Alexanian, *Org. Lett.* **2015**, *17*, 1284.

6 (a) P. Mauleón, R. M. Zeldin, A. Z. González and F. D. Toste, *J. Am. Chem. Soc.* **2009**, *131*, 6348. (b) I. Alonso, B. Trillo, F. López, S. Montserrat, G. Ujaque, L. Castedo, A. Lledós and J. L. Mascareñas, *J. Am. Chem. Soc.* **2009**, *131*, 13020. (c) A. Z. González and F. D. Toste, *Org. Lett.* **2010**, *12*, 200.

7 (a) B. M. Trost, D. R. Fandrick and D. C. Dinh, *J. Am. Chem. Soc.* **2005**, *127*, 14186. (b) J. K. Lam, Y. Schmidt and C. D. Vanderwal, *Org. Lett.* **2012**, *14*, 5566.

8 F. López and J. L. Mascareñas, *Chem. Soc. Rev.* **2014**, *43*, 2904.

9 For selected reviews on the synthesis of allenes, see: (a) *Modern Allene Chemistry*, (Eds.: N. Krause, A. S. K. Hashmi), Vols. 1 and 2, Wiley-VCH, Weinheim, **2004**. (b) S. Yu and S. Ma, *Chem. Commun.* **2011**, *47*, 5384. (c) K. M. Brummond and J. E. DeForrest, *Synthesis* **2007**, 795. (d) R. K. Neff. and D. E. Frantz, *ACS Catal.* **2014**, *4*, 519. (e) J. Ye and S. Ma, *Org. Chem. Front*. **2014**, *1*, 1210. (f) M. Ogasawara, *Tetrahedron: Asymmetry* **2009**, *20*, 259.

10 For selected reports on the synthesis of 1,3-disubstituted allenes, see: (a) J. Kuang and S. Ma, *J. Am. Chem. Soc.* **2010**, *132*, 1786. (b) J. Kuang, H. Luo and S. Ma, *Adv. Synth. Catal.* **2012**, *354*, 933.

11 Crystal data for compound (3a*R**,5*S**,7a*R**,*Z*)-**2c**: $C_{23}H_{31}NO_2S$, *MW* = 385.55, Triclinic, space group P -1, final *R* indices [*I* > 2σ(*I*)], *R*1 = 0.0550, *wR*2 = 0.1388, *R* indices (all data) *R*1 = 0.0764, *wR*2 = 0.1388, *a* = 6.182(5) Å, *b* = 11.371(8) Å, *c* = 13.441(9) Å, *α* = 80.287(13)°, *β* = 84.840(13)°, *γ* =77.986(12)°, *V* = 1002.2(12) Å^3, *T* = 130 K, *Z* = 2, reflections collected/unique 14550/4638 (*R*int = 0.0470), number of observations [> 2σ(*I*)] 4638, parameters: 246. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Centre, CCDC 1528081.

12 Crystal data for compound (3a*R**,7a*S**,*E*)-**3j**: $C_{24}H_{30}O_4$, *MW* = 382.48, Monoclinic, space group P1(21)/c(1), final *R* indices [*I* > 2σ(*I*)], *R*1 = 0.0453, *wR*2 = 0.1060, *R* indices (all data) *R*1 = 0.0781, *wR*2 = 0.1227, *a* =8.9632(10) Å, *b* = 26.894(3) Å, *c* = 9.2793(10) Å, *α* = 90°, *β* = 110.151(2)°, *γ* = 90°, *V* = 2099.9(4) Å^3, *T* = 130 K, *Z* = 4, reflections collected/unique 21185/6552 (*R*int = 0.0373), number of observations [> 2σ(*I*)] 6552, parameters: 256. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Centre, CCDC 1555723.

13 Crystal data for compound (3a*R**,5*R**,7a*R**,*Z*)-**2b**: $C_{18}H_{23}NO_2S$, *MW* = 317.43, monoclinic, space group P 21/c, final *R* indices [*I* > 2σ(*I*)], *R*1 = 0.0486, *wR*2 = 0.1232, *R* indices (all data) *R*1 = 0.0611, *wR*2 = 0.1316, *a* = 10.6868(14) Å, *b* = 20.200(3) Å, *c* = 8.0112(10) Å, *α* = 90°, *β* = 101.747(3)°, *γ* = 90°, *V* = 1693.2(4) Å^3, *T* = 293(2) K, *Z* = 4, reflections collected/unique 9268/2981 (*R*int = 0.0374), number of observations [> 2σ(*I*)] 2981, parameters: 202. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Centre, CCDC 1555722.

Organic Chemistry Frontiers Accepted Manuscript